The impact of the multidisciplinary team in the management of individuals with diabetic foot ulcers: a systematic review

Objective: To assess the impact of the multidisciplinary team in the management of the diabetic foot compared with those who did not receive multidisciplinary care.
Method: A systematic review of the literature was conducted using the following databases: Cumulative Index to Nursing and Allied Health Literature (CINAHL), Medline, Embase and Cochrane Library. The following search terms were used: diabetic foot, multidisciplinary team, patient care team, multidisciplinary care team. Data were extracted using a bespoke data extraction tool and quality appraisal of the studies was undertaken using the EBL Critical Appraisal checklist. Data analysis was undertaken using RevMan with results presented as odds ratio for dichotomous data, or mean difference for continuous data, all with the associated 95% confidence intervals.
Results: The search identified 19 eligible studies. Severity of amputation, death rates and length of hospital stay of clients receiving multidisciplinary team care were improved when compared with those who did not receive multidisciplinary team care. Ulcer healing and quality of life showed an improvement but not all studies explored these outcomes. Only 7 of the 19 articles appraised were found to be of acceptable quality, questioning the generalisability of the results.
Conclusion: From the currently available evidence a positive impact of the multidisciplinary team on diabetic foot outcomes can be seen, but due to the lack of high-quality evidence and substantial heterogeneity in the studies, these results should be interpreted with caution.
Declaration of interest: The authors have no conflicts of interest.

diabetic foot • amputation • multidisciplinary team • ulceration • DFUs

Among the many complications that can occur due to diabetes, foot ulceration is a leading problem.[1] Diabetic foot ulcers (DFUs) occur due to various risk factors associated with diabetes such as sensory neuropathy, peripheral ischaemia and foot deformities.[2] The presence of any of these factors can result in complications that can progress rapidly.[3] When we look at Ireland, for example, where the authors of this review are located, the prevalence of people over 20 years of age living with diabetes in 2005 was estimated to be 4.7% and this figure was predicted to rise to 5.0% in 2015.[4] Within this group it is estimated that there are between 20,470 and 41,020 episodes of diabetic foot ulceration in Ireland at any one time.[5] However, this is not just a national problem isolated to Ireland, but rather a global issue. For example, in some islands in the Caribbean, prevalence rates of diabetes are 10–20%.[6] Furthermore, in the UK one in 20 adults is thought to have diabetes, with UK prevalence rates estimated at 4.45%.[7]

***A. Buggy**,[1] MSc (Wound Management and Tissue Viability), PG Dip, Professional Certificate (Community Nursing), RGN, CNM 2 Integrated Case Manager for Older Persons; **Z. Moore**,[1] PhD, Msc (leadership in Health Professions Education) MSc (Wound Healing and Tissue Repair), FFNMRCSI, PG Dip, Dip First Line Management, RGN, Professor and Head of the School of Nursing and Midwifery
Corresponding author e-mail: aislingbuggy@rcsi.ie
1 Royal College of Surgeons in Ireland, Dublin, Ireland.

Treatment of diabetic foot disease has been found to be the most costly of all diabetic complications.[8] For example, the estimated cost of treating one client as an inpatient with a foot ulcer in Ireland is €30,000.[8] In addition to monetary cost, foot disease is a physical disability and has a huge impact on a person's quality of life.[9] Furthermore, a high risk of limb amputation is associated with the condition, with over 85% of amputations occurring in individuals with active or previous DFUs.[6] Amputation is such a huge problem that it has been reported that a limb is lost to diabetes somewhere in the world every 30 seconds, a statistic that can be lowered significantly if appropriate approaches are provided.[10] A key element suggested in the prevention of diabetic foot disease is multidisciplinary team (MDT) involvement in client care.[11,12] An MDT is considered to be a pinnacle part of diabetic foot care and should consist of a group of health-care professionals each with knowledge of different aspects of the diabetic foot, collaborating together.[13]

The National Institute for Health and Care Excellence (NICE) has developed guidelines on the prevention and management of the diabetic foot.[14] The need for a person with diabetes to be referred to an MDT is repeated throughout these guidelines at all stages of disease progression and for the prevention of the disease. However, the structure and delivery of these teams varies widely across health-care settings.[14,15] In addition, in England it is reported that around one-fifth

© 2017 MA Healthcare ltd

Request Permissions
© MA Healthcare Ltd. Downloaded from magonlinelibrary.com by 155.198.030.043 on June 14, 2017.
Use for licensed purposes only. No other uses without permission. All rights reserved.

Figure 1. PRISMA flow chart

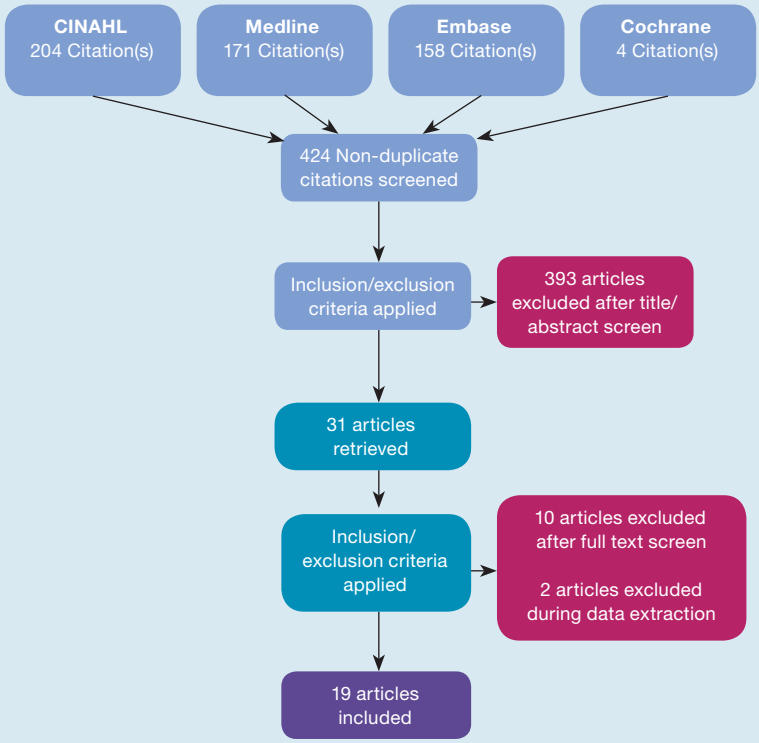


Table 1. Excluded studies

Study	Reason for exclusion
Thewjitcharoen et al. (2014)[20]	No comparison. Only reported what happened after the MDT was established
Van Gils et al.(1999)[21]	Study also looked at non-diabetics also and data not separate for diabetics
Sinkin et al. (2015)[22]	No clear mention of MDT in the article
Ignatyeva et al. (2015)[23]	Excluded as focus was on government paying for MDT clinic. Primary outcomes not addressed
Chen et al. (2011)[24]	Foot care not discussed in study. No primary outcomes reported
Valdés et al. (1999)[25]	Focused on all wounds. No separate data for diabetics to extract
El Sakka et al. (2006)[26]	No comparative group or 'before' data to extract
Krishnan et al. (2008)[27]	No comparative/before data given
Manu et al. (2014)[28]	MDT already in place but delivery of it changed. No clear comparison
Lazzarini et al. (2012)[29]	Explored many quality-improvement strategies. No separate data for MDT effect
Crane and Werber (1999)[30]	Not MDT. Only care from podiatry included

of hospitals that provide diabetes care have no MDT diabetic foot service.[16] The purpose of this systematic review was to explore the delivery of a diabetic foot MDT in greater detail and to identify if its existence leads to more positive diabetic foot outcomes. Thus, this systematic review explored the following question: What is the impact of the MDT in the management of individuals with DFUs?

Methods

PICO

Using the PICO model, the participants of concern were adults aged 18 years or older with type 1 or type 2 diabetes mellitus, cared for in any health-care setting, with an active foot ulcer of any stage. The intervention of interest was the delivery of DFU treatment with an MDT approach, whereupon, the team had to be multidisciplinary health professionals. The comparison of interest was the delivery of care to manage people with DFUs by an individual health professional or using an unstructured, non-collaborative care model. The primary outcome measures were amputation rates, ulcer development and wound healing. With regard to amputation rates this included the level of amputation. Secondary outcome measures were: patients' experiences, mortality and resource use, which included length of hospital stay and cost of care. All English-language studies employing the following designs were included: randomised control trials, controlled clinical trials and comparative pre and post studies.

Search strategy

The following databases were searched over a period from 19 January 2016 to 21 January 2016: Cumulative Index to Nursing and Allied Health Literature (CINAHL), Medline, Embase and Cochrane Library. The following search terms were used: diabetic foot, multidisciplinary team, patient care team, multidisciplinary care team. The variation in terms was determined by the MeSH headings for each individual database. No limit was placed on year of publication. A limitation of English language only was placed on each search. The reference lists of chosen articles were screened but no additional articles were obtained from this screening.

Data collection

Potentially eligible studies were selected by screening titles and abstracts, independently, by two reviewers. Full copies of potentially eligible studies were then obtained and read. Articles were then selected independently, by two reviewers, by applying the predefined inclusion and exclusion criteria.

Data extraction

Further articles were excluded at the data-extraction stage if they did not report on the outcomes required. The data from the included articles were extracted by two reviewers. The information extracted consisted of the first author, study title, year of publication, country where study was carried out, study design, care setting, sample size and findings.

Data analysis and synthesis

Where suitable studies were analysed using Revman 5.3.[17] The results of the remaining studies were analysed descriptively. Results are presented as odds ratio (OR) for dichotomous data with the associated 95%

© 2017 MA Healthcare Ltd

Request Permissions
© MA Healthcare Ltd. Downloaded from magonlinelibrary.com by 155.198.030.043 on June 14, 2017.
Use for licensed purposes only. No other uses without permission. All rights reserved.

confidence interval (CI). Results for continuous data are presented as mean difference (MD) with the associated 95%. Statistical significance was set at p≤0.05. Where appropriate, a meta-analysis was undertaken of the results. Where meta-analysis was not possible, a narrative synthesis of the studies was undertaken consisting of a summary of each study including an outline of the author, sample size, results and a synopsis of these results.

Quality appraisal

Quality appraisal was undertaken using the EBL Critical Appraisal checklist.[18] For systematic reviews the EBL checklist was not suitable so the PRISMA 2009 Checklist was employed.[19] Each stage of the research process was validated, independently, by a second reviewer. In the case of disagreement, the plan was for consensus to be reached through discussion.

Results

The results from the literature search are shown in the PRISMA flow chart (Fig 1). A total of 537 articles were obtained for screening. Duplicates were then removed leaving a total of 424 articles. The titles and abstracts of the remaining articles were screened. Following this, 31 articles remained for further screening, ten articles of which were excluded (Table 1), leaving 21 articles.[20–30] A further two were excluded at the data extraction stage as they did not fit the inclusion criteria (Table 1). From this process 19 articles remained and these formed the basis of the review.

Geographical location

The geographical location of the studies varied, three were conducted in Turkey,[31–33] two in Taiwan,[34,35] two in Spain,[36,37] two in the US,[38,39] one in the UK,[40] one in Singapore,[41] one in Belgium,[42] one in Thailand,[43] one in Lithuania,[44] one in Ireland,[45] one in Israel,[46] one in Denmark,[47] and one in Sweden.[48] There was one systematic review.[49]

Care setting

All studies were based in a hospital care setting. The majority focused on an acute care setting.[31–33,35–36,38–43,45–48] Three focused on an outpatient department or hospital-based clinic,[34,44,49] and one looked at data from both an acute and outpatient department setting.[37]

Participants and sample size

All participants had a diagnosis of diabetes with complications affecting the feet. This included past and/or present foot ulceration. In one study data for diabetic patients and non-diabetic patients who attended the MDT was provided.[36] These were presented separately for the diabetic participants and so this study was eligible for inclusion.[36] The smallest reported sample size was 66.[32] No mean sample size could be obtained as some studies did not report their full sample size,[35,36,39,40] or only reported the experimental sample size and not that of the control.[45]

Table 2. Makeup of multidisciplinary teams (MDTs)

Study	MDT members
Alexandrescu et al. (2009)[42]	Diabetologist, vascular consultant, radiotherapy, plastics, psychologists, infectious diseases, nurses, orthotics and general practitioner
Armstrong et al. (2012)[39]	Not listed
Askoy et al. (2004)[32]	Not listed
Aydin et al. (2010)[31]	Not listed
Cahn et al. (2014)[46]	Orthopaedics, endocrinologist and nurse
Chiu et al. (2010)[34]	Plastic surgeon and diabetic foot ulcer treatment programme. Members not listed
Dargis et al. (1999)[44]	Diabetologist, rehabilitation physician, podiatrist, orthopaedic surgeon
Hedetoft et al. (2009)[47]	Diabetes specialist, orthopaedics, podiatrist and nurse.
Hsu et al. (2015)[35]	Vascular surgeon, general surgery, plastics, diabetologist, physiotherapist, occupational therapist, orthopedics, dermatology, microbiology, radiology, home-care team and general practitioner
Larsson et al. (1995)[48]	Diabetologist, orthopaedics, diabetes nurse, podiatrist, orthotist and vascular surgeon
Martinez-Gòmez et al. (2014)[37]	Diabetologist, emergency medicine, anaesthetics, surgery, infectious diseases, pharmacy, medical and rehabilitation, psychiatry, medical, general practitioner, nurse, social work
Meltzer et al. (2002)[38]	Podiatrist, physician, plastic surgeon, vascular surgeon, orthotist, social work
Nason et al. (2013)[45]	Vascular surgeon, endocrinologist, podiatrist, orthotist, tissue viability nurse
Nather et al. (2010)[41]	Endocrinologist, orthopaedics, infectious diseases, vascular surgeon, podiatrist, tissue viability nurse, foot care and screening nurse, case manager
Quinlivan et al. (2014)[49]	Systematic review—not listed
Rerkasem et al. (2009)[43]	'Diabetes foot protocol', members not listed
Rubio et al. (2014)[36]	Not listed
Williams et al. (2012)[40]	Vascular consultant, vascular nurse specialist, two podiatrists, lower limb wound nurse, orthotist
Yesil et al. (2009)[33]	Endocrinologist, orthopaedics, plastics, vascular consultant, infectious diseases, rehab, radiologist, diabetes nurse, wound nurse

Study designs

The studies' designs included 11 retrospective studies,[31–32,35–40,42,45–46,48] 4 prospective studies,[33,40,41,44] two controlled clinical trials,[43,47] one was a randomised control trial[34] and one a systematic review.[49]

Interventions and follow-up

All studies evaluated the effect of an MDT on diabetic foot care compared with no MDT. One study looked at care before a team was established, after a team was established and then also included later data from when this MDT had commenced a foot clinic.[37] Each study had a different combination of members in the team or did not list the exact members. The reported makeup of each team can be found in Table 2.

© 2017 MA Healthcare ltd

Request Permissions

© MA Healthcare Ltd. Downloaded from magonlinelibrary.com by 155.198.030.043 on June 14, 2017.
Use for licensed purposes only. No other uses without permission. All rights reserved.

Table 3. EBL checklist results

Study author (date)	EBL score %
Alexandrescu et al. (2009)[42]	15/25 = 60% not valid
Armstrong et al. (2012) [39]	15/25 = 60% not valid
Askoy et al. (2004)[32]	15/25 = 60% not valid
Aydin et al. (2010)[31]	20/26 = 77% valid
Cahn et al. (2014)[46]	20/24 = 83% valid
Chiu et al. (2010)[34]	23/26 = 88% valid
Dargis et al. (1999)[44]	21/26 = 81% valid
Hedetoft et al. (2009)[47]	21/26 = 81% valid
Hsu et al. (2015)[35]	18/25 = 72% not valid
Larsson et al. (1995)[48]	18/25 = 72% not valid
Martinez-Gòmez et al. (2014)[37]	16/25 = 64% not valid
Meltzer et al. (2002)[38]	16/25 = 64% not valid
Nason et al. (2013)[45]	16/26 = 62% not valid
Nather et al. (2010)[41]	20/26 = 77% valid
Rerkasem et al. (2009)[43]	19/24 = 79% valid
Rubio et al. (2014)[36]	18/25 = 72% not valid
Williams et al. (2012)[40]	14/24 = 58% not valid
Yesil et al. (2009)[33]	15/25 = 60% not valid

Quality appraisal of included studies

Quality appraisal was carried out on 18 of the studies using the EBL critical appraisal checklist.[18] Using this checklist, the study is shown to be valid if the proportion of 'yes' answers is ≥75%, or if 'no' or 'unclear' answers are ≤25%. The overall validities are presented in Table 3. The mean overall score was 71%, with the lowest being 58%[40] and the highest being 88%.[34] For the included systematic review[49] the PRISMA checklist, which contains a list of 27 prompts, was used for quality assessment. These prompts are considered to be critical for the development of a transparent and valid systematic review.[51] Analysing the systematic review we found an answer of yes to 12 questions and no to 15 meaning there are flaws in their reporting and in some incidences a lack of clarity.

Primary outcomes amputation rates

Amputation rates were reported in all of the studies. This incorporated either one or a combination of total amputations, major amputations, minor amputations and re-amputation rates. Meta-analysis could not be undertaken for total amputation rates due to a substantial heterogeneity (I-squared=69%, p=0.007). If I-squared is >30%, or there is a low p-value (<0.10) in the chi-squared test for heterogeneity then it is not appropriate to undertake meta-analysis.[50] Therefore total amputation rates had to be analysed separately with a combination of statistical and narrative analyses.

RevMan analysis of one study showed a statistically significant difference in favour of the experiment group when compared with the control group.[35] Those in the experimental group were almost five times less likely to have a lower extremity amputation than those in the control group (MD⁻4.97, 95% CI ⁻5.21–⁻4.73; p≤0.00001) (Fig 2).

A further three studies found a significant statistical difference in favour of the experimental group when comparing total amputation rates pre and post MDT care,[34,37,38] [OR 0.35; 95% CI: 0.23–0.53; p<0.00001][34] (Fig 3), [OR 0.35; 95% CI: 0.21–0.60; p=0.0001][38] (Fig 4), [OR 0.65; 95% CI: 0.47–0.90; p=0.01037] (Fig 5).

RevMan analyses of four further studies did not identify statistically significant differences in total amputation rates between the control and experimental groups[32,33,43,46] [OR 1.86;95% CI: 0.34–10.12; p=0.47][43] (Fig 6), [OR 1.12; 95% CI: 0.62–2.03; p=0.71][32] (Fig 7), [OR 0.60;95% CI: 0.34–1.06; p=0.08][46] (Fig 8), [OR 0.85; 95% CI 0.58–1.26; p=0.43][33] (Fig 9).

For those studies that were analysed narratively, differences between MDT care and no MDT care were also seen. Of these one study reported a 36.7% and 39.4% total amputation rate before the MDT was established and an amputation rate of 21.1% after the team was established.[31] Another study found a statistically significant difference (p<0.001) when analysing amputation rates before an MDT, after an MDT was established and after this MDT commenced a foot clinic.[37] The study reports a 60.8% amputation rate in the control period, 50.2% amputations post MDT establishment and 42.8% amputations after this MDT commenced a foot clinic.

A high to low amputation ratio of 0.35 to 0.27 was reported in a further study, arising due to an increase in low level midfoot amputations (8.2% versus 26.1%, p<0.0001).[39] A statistically significant difference was documented in one study, in favour of the experimental group with 17 amputations documented pre the MDT and 10 post (p=0.040).[42] A final study found no statistically significant differences when evaluating total amputation rate pre and post an MDT establishment with 4.0 [95% CI: 2.6–5.5] pre MDT and 6.1 [95% CI: 4.9–7.2] post MDT.[36] No sample sizes were given for this study in the text.

Another study reported that the total number of annual amputations decreased by 49% after establishing a diabetic MDT.[48] This study did not separate sample size for the control and experiment. Finally, the included systematic review reports a statistically significant decrease in amputation rates in five out of the seven included studies when an MDT approach was compared with standard care.[49]

Primary outcomes: major and minor amputations

Of the studies nine reported major amputations,[33,37,40–42,44–46] and eight minor amputations.[33,36,37,40,44–47] Meta-analysis of the results of five of these studies for major amputations, found that there was a 37% reduction in the odds of having an amputation in the experimental group compared with the control group [OR .63; 95% CI: 0.48–0.82; p=0.0006) (Fig 10).[42,46,44,47,37,33]

The studies that were not suitable for the

© 2017 MA Healthcare ltd

Request Permissions

© MA Healthcare Ltd. Downloaded from magonlinelibrary.com by 155.198.030.043 on June 14, 2017.
Use for licensed purposes only. No other uses without permission. All rights reserved.

Fig 2. Lower-extremity amputations

	Experimental			Control				Mean difference
Study or subgroup	**Mean**	**SD**	**Total**	**Mean**	**SD**	**Total**	**Weight**	**IV, Fixed, 95% CI**
Hsu et al. (2015)	3.02	1.26	637	7.99	1.55	186	100.0%	-4.97 -5.21, -4.73]
Total (95% CI)			**637**			**186**	**100.0%**	**-4.97 -5.21, -4.73]**
Heterogeneity: Not applicable								
Test for overall effect: Z=40.04 (p<0.00001)								

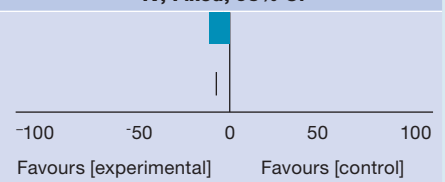


Fig 3. Total amputations: A

	Experimental		Control			Odds ratio
Study or subgroup	**Events**	**Total**	**Events**	**Total**	**Weight**	**M-H, Fixed, 95% CI**
Chiu et al. (2010)	34	350	91	386	100.0%	0.35 [0.23, 0.53]
Total (95% CI)		**350**		**386**	**100.0%**	**0.35 [0.23, 0.53]**
Total events	34		91			
Heterogeneity: Not applicable						
Test for overall effect: Z=4.86 (p<0.00001)						

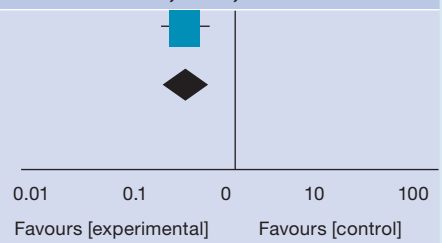
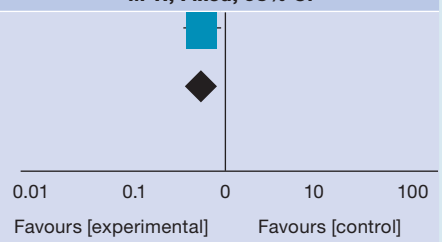


Fig 4. Total amputations: B

	Experimental		Control			Odds ratio
Study or subgroup	**Events**	**Total**	**Events**	**Total**	**Weight**	**M-H, Fixed, 95% CI**
Meltzer et al. (2002)	45	116	76	118	100.0%	0.35 [0.21, 0.60]
Total (95% CI)		**116**		**118**	**100.0%**	**0.35 [0.21, 0.60]**
Total events	45		76			
Heterogeneity: Not applicable						
Test for overall effect: Z=3.88 (p=0.00001)						

Fig 5. Total amputations: C

	Experimental		Control			Odds ratio
Study or subgroup	**Events**	**Total**	**Events**	**Total**	**Weight**	**M-H, Fixed, 95% CI**
Martinez-Gòmez et al. (2014)	220	438	138	227	100.0%	0.65 [0.47, 0.90]
Total (95% CI)		**438**		**227**	**100.0%**	**0.65 [0.47, 0.90]**
Total events	220		138			
Heterogeneity: Not applicable						
Test for overall effect: Z=2.58 (p=0.010)						



meta-analysis also reported a statistical difference reduction in major amputations in favour of the experimental groups also (p<0.0005[41] and p<0.001[37]). In one other study a total of eight major amputations were seen in the control and three in the experiment group.[45] No specific figures were given and further analysis of this data could not be undertaken as no sample size was given for the control group. Similarly in one other study, 41 major amputations took place in the control group and 27 in the experimental group after an MDT was established.[40] However, no sample size was provided so no further anaysis was appropriate. Another study reported that major amputations decreased from 78% previous to the MDT to 16.1% after its implementation (3.6/100000 inhabitants, p=0.001).[48]

Meta-analysis of the data from four studies, pertaining

© 2017 MA Healthcare ltd

Request Permissions
© MA Healthcare Ltd. Downloaded from magonlinelibrary.com by 155.198.030.043 on June 14, 2017.
Use for licensed purposes only. No other uses without permission. All rights reserved.

Fig 6. Total amputations: D

	Experimental		Control			Odds ratio
Study or subgroup	**Events**	**Total**	**Events**	**Total**	**Weight**	**M-H, Fixed, 95% CI**
Rerkasem et al. (2009)	5	56	2	40	100.0%	1.86 [0.34, 10.12]
Total (95% CI)		**56**		**40**	**100.0%**	**1.86 [0.34, 10.12]**
Total events	5		2			
Heterogeneity: Not applicable						
Test for overall effect: Z=0.72 (p=0.47)						

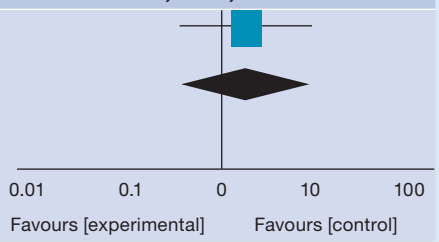
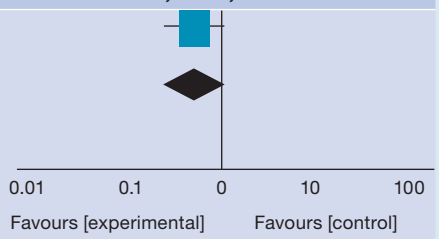


Fig 7. Total amputations: E

	Experimental		Control			Odds ratio
Study or subgroup	**Events**	**Total**	**Events**	**Total**	**Weight**	**M-H, Fixed, 95% CI**
Askoy et al. (2004)	26	66	54	147	100.0%	1.12 [0.62, 2.03]
Total (95% CI)		**66**		**147**	**100.0%**	**1.12 [0.62, 2.03]**
Total events	26		54			
Heterogeneity: Not applicable						
Test for overall effect: Z=0.37 (p=0.71)						

Fig 8. Total amputations: F

	Experimental		Control			Odds ratio
Study or subgroup	**Events**	**Total**	**Events**	**Total**	**Weight**	**M-H, Fixed, 95% CI**
Cahn et al. (2014)	48	101	56	93	100.0%	0.60 [0.34, 1.06]
Total (95% CI)		**101**		**93**	**100.0%**	**0.60 [0.34, 1.06]**
Total events	48		56			
Heterogeneity: Not applicable						
Test for overall effect: Z=1.77 (p=0.08)						



to minor amputations, did not find a statistically significant difference between the study groups [OR 0.96, 95% CI 0.75 to 1.22; p=0.73], (Fig 11).[33,37,44,46] One study that was not suitable for the meta-analysis reported an increase in minor amputations from 81 to 101 when an MDT was established,[40] however, no sample size was provided for this study. A further study noted no change in minor amputation rates, where four below-knee amputations were performed in both the control and experimental groups.[45] Similarly, in an additional study 56 minor amputations were performed in the control group and 44 for the experiment.[47] Further analysis using RevMan was not possible as the events for the experimental group exceeded the sample size of 28, meaning some participants had more than one minor amputation. A further study reported that minor amputations increased from 28% before the MDT to 53%.[48] Only one study noted a statistically significant difference in minor amputation rate, in favour of the experimental group;[36] intervention, 5.0 (2.3–7.8), control, 5.7 (3.9–7.5) p=0.732.

Primary outcomes: re-amputation

Re-amputation rate was documented in two studies.[34,48] In both a statistically significant difference was seen in favour of the experimental group. A 59% reduction in the odds of having a re-amputation was found in one study [OR 0.41; 95% CI 0.20 to 0.85; p=0.02] (Fig 12),[34] and a decrease from 36% to 22% (p<0.05) was reported in the second, however, the specific data were not presented in the study.[48]

One study also provided data for Syme amputations (an amputation through the ankle joint), trans-metatarsal, and ray and trans-phalangeal amputations[32] Syme amputations accounted for 7.4% of amputations in the experimental group and 8% in the control. Trans-metatarsal accounted for 3.7% in

© 2017 MA Healthcare ltd

Request Permissions
© MA Healthcare Ltd. Downloaded from magonlinelibrary.com by 155.198.030.043 on June 14, 2017.
Use for licensed purposes only. No other uses without permission. All rights reserved.

Fig 9. Total amputations: G

	Experimental		Control			Odds ratio
Study or subgroup	**Events**	**Total**	**Events**	**Total**	**Weight**	**M-H, Fixed, 95% CI**
Yesil et al. (2009)	159	437	55	137	100.0%	0.85 [0.58, 1.26]
Total (95% CI)		**437**		**137**	**100.0%**	**0.85 [0.58, 1.26]**
Total events	159		55			
Heterogeneity: Not applicable						
Test for overall effect: Z=0.79 (p=0.43)						

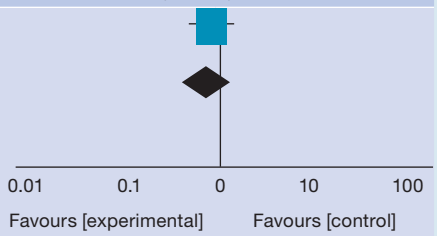


Fig 10. Major amputations

	Experimental		Control			Odds ratio
Study or subgroup	**Events**	**Total**	**Events**	**Total**	**Weight**	**M-H, Fixed, 95% CI**
Alexandrescu et al. (2009)	0	0	0	0		Not estimable
Cahn et al. (2014)	19	101	30	93	18.8%	0.49 [0.25, 0.94]
Dargis et al. (1999)	1	56	4	89	2.2%	0.39 [0.04, 3.55]
Hedetoft et al. (2009)	14	28	28	60	6.6%	1.14 [0.47, 2.80]
Martinez-Gòmez et al. (2014)	79	438	56	227	44.8%	0.67 [0.46, 0.99]
Yesil et al. (2009)	55	437	28	137	27.6%	0.56 [0.34, 0.93]
Total (95% CI)		**1060**		**606**	**100.0%**	**0.63 [0.48, 0.82]**
Total events	168		146			
Heterogeneity: chi-squared=2.78, df=4 (p=0.600; I-squared=0%)						
Test for overall effect: Z=3.42 (p=0.0006)						

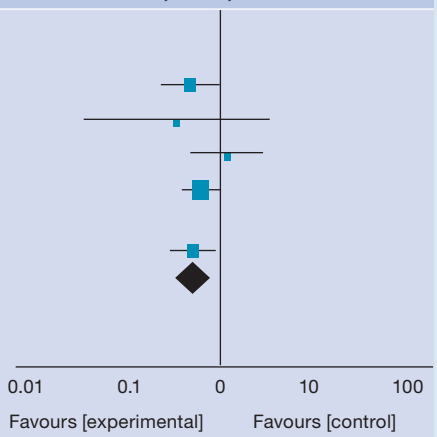
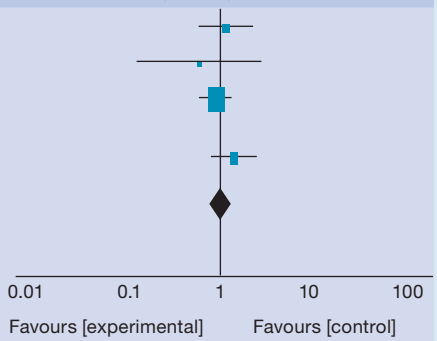


Fig 11. Minor amputations

	Experimental		Control			Odds ratio
Study or subgroup	**Events**	**Total**	**Events**	**Total**	**Weight**	**M-H, Fixed, 95% CI**
Cahn et al. (2014)	29	101	26	93	14.9%	1.04 [0.56, 1.94]
Dargis et al. (1999)	3	56	8	89	4.5%	0.57 [0.15, 2.26]
Martinez-Gòmez et al. (2014)	141	438	82	227	56.4%	0.84 [0.60, 1.18]
Yesil et al. (2009)	103	437	27	137	24.2%	1.26 [0.78, 2.02]
Total (95% CI)		**1032**		**546**	**100.0%**	**0.96 [0.75, 1.22]**
Total events	276		143			
Heterogeneity: chi-squared=2.44, df=3 (p=0.49; I-squared=0%						
Test for overall effect: Z=0.34 (p=0.73)						



the experimental group and 0% for the control. There were no ray amputations in the experimental group, but 35% occurrence in the control group. Transphalangeal accounted for 25.9% of the amputations in the experimental group and 0% in the control. Non-specified amputations classified as 'others', were not undertaken in the experimental group, however, they were undertaken in 5% of cases within the control group.

Primary outcomes: ulcer healing

Ulcer healing was documented differently in the studies. Recurrent foot ulcer rate at a two-year follow-up appointment was assessed in one study which found a statistically significant difference before and after.[44] There was a 69% reduction in the odds of having a recurrent foot ulcer in the experimental group compared with the control group [OR 0.31; 95% CI 0.15–0.63; p=0.001] (Fig 13).

© 2017 MA Healthcare ltd

Request Permissions

© MA Healthcare Ltd. Downloaded from magonlinelibrary.com by 155.198.030.043 on June 14, 2017.
Use for licensed purposes only. No other uses without permission. All rights reserved.

Fig 12. Re-amputation rate

	Experimental		Control			Odds ratio
Study or subgroup	**Events**	**Total**	**Events**	**Total**	**Weight**	**M-H, Fixed, 95% CI**
Chiu et al. (2010)	11	350	28	386	100.0%	0.41 [0.20, 0.85]
Total (95% CI)		**350**		**386**	**100.0%**	**0.41 [0.20, 0.85]**
Total events	11		28			
Heterogeneity: Not applicable						
Test for overall effect: Z=2.42 (p=0.02)						

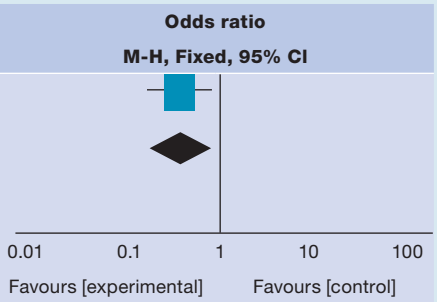
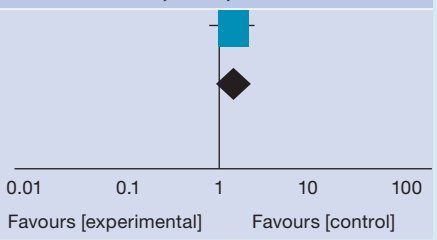


Fig 13. Recurrent foot ulcer rate at two-year follow-up

	Experimental		Control			Odds ratio
Study or subgroup	**Events**	**Total**	**Events**	**Total**	**Weight**	**M-H, Fixed, 95% CI**
Dargis et al. (1999)	17	56	52	89	100.0%	0.31 [0.15, 0.63]
Total (95% CI)		**56**		**89**	**100.0%**	**0.31 [0.15, 0.63]**
Total events	11		28			
Heterogeneity: Not applicable						
Test for overall effect: Z=3.24 (p=0.001)						

Fig 14. Healed ulcers (without amputation)

	Experimental		Control			Odds ratio
Study or subgroup	**Events**	**Total**	**Events**	**Total**	**Weight**	**M-H, Fixed, 95% CI**
Yesil et al. (2009)	220	437	60	137	100.0%	1.30 [0.88, 1.91]
Total (95% CI)		**437**		**137**	**100.0%**	**1.30 [0.88, 1.91]**
Total events	220		60			
Heterogeneity: Not applicable						
Test for overall effect: Z=1.34 (p=0.18)						



A further study looked at ulcers that healed without requiring amputation in addition to the rate of unhealed ulcers.[33] RevMan analysis found no statistically significant difference between the groups [OR 1.30; 95% CI: 0.88–1.91, p=0.18] (Fig 14) and [OR 0.82; 95% CI: 0.48–1.39, p=0.45] (Fig 15).

Secondary outcomes: length of stay in hospital

Meta-analysis for outcomes related to length of hospital stay could not be undertaken due to a substantial heterogeneity among the studies (I-squared=96%). The studies were, therefore, analysed individually using RevMan. Two studies showed statistically significant differences in length of hospital stay in favour of the experimental groups ([MD ⁻5.80 95% CI: ⁻7.69– ⁻3.91, p<0.00001][33] (Fig 16), [MD ⁻12.48 95% CI: ⁻17.62–⁻7.34, p <0.00001][34] (Fig 17).

In two other studies there was statistically significant differences observed between the study groups and length of stay in hospital, [OR 0.20, 95% CI ⁻0.47– 0.87; p=0.56;[37] OR 0.80; 95% CI: 0.42–1.53, p=0.50][46] (Fig 18 and 19).

Length of hospital stay was reported in one study as 12 days for the experiment and 15 days for the control.[45] The authors also report a reduction in admissions from 131 people in the control group to 120 people in the experiment group. A statistically significant difference in length of hospital stay, in favour of the experimental group, was reported in another study (p=0.0005).[41] This study presented the average length of hospital days per year of the study, before team formation in 2002, after team formation, 2003–2007. The results were as follows; 2002, 20.36, 2003, 19.03, 2004, 13.74, 2005, 10.81, 2006, 11.67, 2007, 12.20. This study also reports a non-statistically significant difference in the readmission rate among the study groups (p>0.05), with readmission percentage rate reported for each year of the study as: 2002, 13.11%, 2003, 7.14%, 2004, 6.76%, 2005, 7.22%,

© 2017 MA Healthcare ltd

Request Permissions
© MA Healthcare Ltd. Downloaded from magonlinelibrary.com by 155.198.030.043 on June 14, 2017.
Use for licensed purposes only. No other uses without permission. All rights reserved.

Fig 15. Unhealed ulcers

	Experimental		Control			Odds ratio
Study or subgroup	**Events**	**Total**	**Events**	**Total**	**Weight**	**M-H, Fixed, 95% CI**
Yesil et al. (2009)	59	437	22	137	100.0%	0.82 [0.48, 1.39]
Total (95% CI)		**437**		**137**	**100.0%**	**0.82 [0.48, 1.39]**
Total events	59		22			
Heterogeneity: Not applicable						
Test for overall effect: Z=0.75 (p=0.45)						

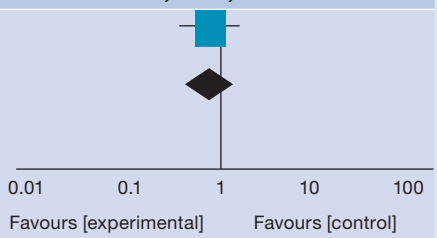


Fig 16. Average days in hospital: A

	Experimental			Control				Mean difference
Study or subgroup	**Mean**	**SD**	**Total**	**Mean**	**SD**	**Total**	**Weight**	**IV, Fixed, 95% CI**
Chiu et al (2010)	23.5	5.8	350	29.3	17.9	386	100.0%	-5.80 [-7.69, -3.91]
Total (95% CI)			**350**			**386**	**100.0%**	**-5.80 [-7.69, -3.91]**
Heterogeneity: Not applicable								
Test for overall effect: Z=6.03 (p<0.00001)								

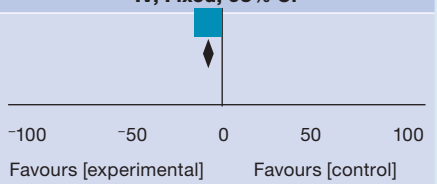


Fig 17. Average days in hospital: B

	Experimental			Control				Mean difference
Study or subgroup	**Mean**	**SD**	**Total**	**Mean**	**SD**	**Total**	**Weight**	**IV, Fixed, 95% CI**
Yesil et al (2009)	26.99	21.27	437	39.47	28.29	137	100.0%	−12.48 [−17.62, −7.34]
Total (95% CI)			**437**			**137**	**100.0%**	**−12.48 [−17.62, −7.34]**
Heterogeneity: Not applicable								
Test for overall effect: Z=4.76 (p<0.00001)								

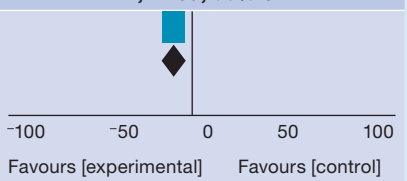


Fig 18. Average days in hospital: C

	Experimental			Control				Mean difference
Study or subgroup	**Mean**	**SD**	**Total**	**Mean**	**SD**	**Total**	**Weight**	**IV, Fixed, 95% CI**
Martinez-Gòmez et al (2014)	11.4	4.4	438	11.2	4.1	227	100.0%	0.20 [−0.47, 0.87]
Total (95% CI)			**438**			**227**	**100.0%**	**0.20 [−0.47, 0.87]**
Heterogeneity: Not applicable								
Test for overall effect: Z=0.58 (p=0.56)								

2006, 5.34%, 2007, 8.26%. A further study looked at average length of hospital stay days between the years 2004 and 2009.[40] The MDT was established in 2006. The results presented from 2004 to 2009 were: 2004, 16, 2005, 18, 2006, 17, 2007, 13, 2008, 14, and 2009, 15.5 days, respectively.

Secondary outcomes: hospitalisation

One study reported on the number of diabetics that required hospitalisation and found a statistically significant difference between the groups[48] [OR 0.16; 95% CI: 0.11–0.24, p=0.00001] (Fig 20).[48] This shows that there was a 69% reduction in the odds of the experimental group being admitted into hospital.

Secondary outcomes: cost of care

The cost of care was reported in three studies.[41,43,45] A statistically significant difference was reported in the first study in favour of the experimental group, which found a difference in cost of US$1127.02 for those involved with the MDT foot clinic versus $1824.58 for those receiving standard care (p<0.02).[43]

© 2017 MA Healthcare ltd

Request Permissions
© MA Healthcare Ltd. Downloaded from magonlinelibrary.com by 155.198.030.043 on June 14, 2017.
Use for licensed purposes only. No other uses without permission. All rights reserved.

Fig 19. Average days in hospital: D

	Experimental		Control			Odds ratio
Study or subgroup	**Events**	**Total**	**Events**	**Total**	**Weight**	**M-H, Fixed, 95% CI**
Cahn et al (2014)	24	101	26	93	100.0%	0.80 [0.42, 1.53]
Total (95% CI)		**101**		**93**	**100.0%**	**0.80 [0.42, 1.53]**
Total events	24		26			
Heterogeneity: Not applicable						
Test for overall effect: Z=0.67 (p=0.50)						

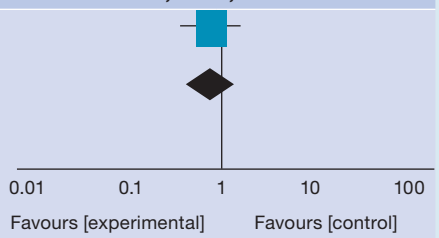


Fig 20. Diabetic foot patients needing hospitalisation

	Experimental		Control			Odds ratio
Study or subgroup	**Events**	**Total**	**Events**	**Total**	**Weight**	**M-H, Fixed, 95% CI**
Hsu et al (2015)	269	637	152	186	100.0%	0.16 [0.11, 0.24]
Total (95% CI)		**637**		**186**	**100.0%**	**0.16 [0.11, 0.24]**
Total events	269		152			
Heterogeneity: Not applicable						
Test for overall effect: Z=8.79 (p<0.00001)						

The second study found a non-statistically significant difference between the groups (p>0.05).[41] This study reported a yearly mean cost per patient as follows: 2002, $8847.17, 2003, $9935.59, 2004, $7659.55, 2005, $6195.77, 2006, $6320.19, 2007, $8383.79. The MDT in the study formed in 2003. Cost is reported in the third study as €2,465,766 for the control and €2,237,641 for the experiment group.[45] However, the sample size for the control group was not reported so more detailed analysis of these figures was not possible.

Secondary outcomes: quality of life

Patients' experiences were explored in one study using a quality of life questionnaire.[43] The results showed a statistically significant difference between the groups, with there being an increase in quality of life found in the experimental group [MD: 8.70; 95% CI 1.07–16.33; p=0.03] (Fig 21)

Secondary outcomes: mortality

Inpatient participant deaths were assessed in two studies.[37,46] When analysed together there was a statistically significant difference between the groups [OR .54; 95% CI: 0.27 to 1.07, p=0.08]. This shows that there is a 46% reduction in the odds of a participant in the experimental group dying (Fig 22.).

Discussion

The multidisciplinary team

The disciplines that made up the MDT differed throughout the articles. The majority contained a variation of consultants, nurse specialists, podiatrists and orthotics. This is comparable with the National Diabetes Programme guidance on the model of care for the diabetic foot, which recommends that diabetic foot care teams should consist of consultants, diabetic nurse specialists and podiatrists.[52] Some of the studies did not discuss the makeup of the MDT,[31,32,34,36,43] leaving it unclear whether the team consisted of the appropriate set of members as recommended in the guidelines. Also, one study described two consultants, orthopaedic and endocrinology, coming together to form a MDT and six months later including a nurse.[46] It could be argued that with two consultants this was not a MDT and even when the nurse was included it did not make up the best recommended team.[13,52] The large variation in the makeup of the MDT seen throughout the included studies leaves it unclear as to if the appropriate health-care professionals were collaborating in all cases (Table 2).[52] In addition, it makes it difficult to compare the results of the 19 studies with each other to draw conclusions when no team had the exact same set of skills.

The delivery settings of the teams were predominately hospital-based inpatient but some also included outpatient clinics. It has previously been documented that the setting can vary once the appropriate health professionals are collaborating and sharing the care.[13] Research containing similar or identical team make-ups would help to further determine if it is in fact the MDT that has an impact on diabetic foot care or if it is the collective efforts of two or more disciplines that works.

© 2017 MA Healthcare ltd

Request Permissions
© MA Healthcare Ltd. Downloaded from magonlinelibrary.com by 155.198.030.043 on June 14, 2017.
Use for licensed purposes only. No other uses without permission. All rights reserved.

Fig 21. Quality of life

	Experimental			Control				Mean difference
Study or subgroup	**Mean**	**SD**	**Total**	**Mean**	**SD**	**Total**	**Weight**	**IV, Fixed, 95% CI**
Rerkasem et al. (2009)	54.7	21.6	56	46	16.5	40	100.0%	8.70 [1.07, 16.33]
Total (95% CI)			**56**			**40**	**100.0%**	**8.70 [1.07, 16.33]**
Heterogeneity: Not applicable								
Test for overall effect: Z = 2.24 (P = 0.03)								

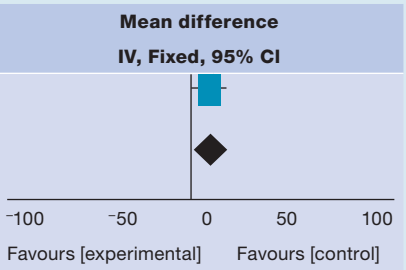
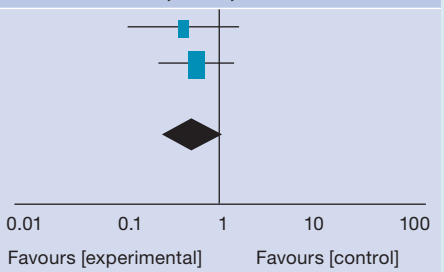


Fig 22. Mortality

	Experimental		Control			Odds ratio
Study or subgroup	**Events**	**Total**	**Events**	**Total**	**Weight**	**M-H, Fixed, 95% CI**
Cahn et al (2014)	4	101	8	93	36.3%	0.44 [0.13, 1.51]
Martinez-Gòmez et al (2014)	13	438	11	227	63.7%	0.60 [0.26, 1.36]
Total (95% CI)		**539**		**320**	**100.0%**	**0.54 [0.27, 1.07]**
Total events	17		19			
Heterogeneity: chi-sqiared=0.17, df=1 (p=0.68); I-squared=0%						
Test for overall effect: Z=1.77 (p=0.08)						



The effect of the MDT on amputation rates

The ultimate aim in the care of the diabetic foot is to avoid ulceration and resulting amputation.[9] This concept was addressed in all 19 of the included studies. Much of the data were collected by analysing patients' notes or databases, retrospectively.[31,32,35–39,42,46,48] This must be recognised as a limitation of these studies as they were relying on the relevant documentation for analysis having been completed accurately. Furthermore, there is a risk that all amputations recorded may not be done so in a way that allows for a collective analysis. This was obvious in some of the studies, given the substantial heterogeneity noted when assessing suitability for meta-analysis.

Total amputations were addressed in 15 of the studies. Of these studies nine showed positive results and six showed negative results. However, when these results were analysed two of the studies that showed positive results and two negative results, others failed to report separate control and experiment sample sizes, which weakens the interpretation of these results.[31,36,39,48] In addition, the included systematic review, which showed a positive impact of MDTs, did not provide any figures or data to demonstrate where the results came from, making these results impossible to analyse.[49]

It has been documented that when effective measures are put in place the severity of a diabetic foot complication can be reduced.[53] Major amputation results were reported in nine studies. When the results of both meta-analyses and narrative analysis were explored, all reported a statistically significant reduction in major amputations after the implementation of an MDT. However, again, two of these did not report a clear sample size leaving the relevance of those results unclear.[40,45]

Minor amputation results were reported in nine studies; seven of the studies favoured the control group and one showed no change in amputation rate.[45] Only one of the nine studies found a statistically significant reduction in minor amputations in the experiment group. On the surface this may appear to be a negative impact of the MDT but it could be argued that this was a favourable outcome as the results would indicate the severity of amputation decreases once an individual is under the care of an MDT. However, what is meant by major and minor amputations is not explained in the studies so what they entail must just be presumed. Only one study divided the results into very specific amputations.[32] More studies like this would be beneficial to understand the precise impact on amputation severity.

Re-amputation rates were reduced in both the studies reporting this outcome.[34,48] However, neither study reports on the follow-up care after the initial amputation. It was noted that early intervention and the type of surgery performed was a contributing factor to lower re-amputation rates in the experiment group in one study.[34] The other does not place much emphasis on this in the results as some of the ulcers followed in the study had yet to heal and potentially could result in re-amputation at a later time.[48] Therefore, for the purpose of this review no conclusion can be made regarding the impact of the MDT on re-amputation rates.

© 2017 MA Healthcare ltd

Request Permissions
© MA Healthcare Ltd. Downloaded from magonlinelibrary.com by 155.198.030.043 on June 14, 2017.
Use for licensed purposes only. No other uses without permission. All rights reserved.

MDT and ulcer healing

Ulcer healing is also cited as a primary goal of the diabetic MDT.[9] This was only explored in two of the included studies.[33,44] Although one of these studies found a statistically significant difference in favour of the experimental group,[44] no firm conclusions can be made in relation to ulcer healing due to the lack of research in this area. More robust research in the area of ulcer healing, progression and in fact also ulcer prevention—which was not documented in any study—is essential to determine the impact of the multidisciplinary foot care team on this primary goal.

MDT and resource use.

The treatment of diabetic foot disease is costly and often results in frequent hospitalisation.[52] Hospitalisation was reported in seven studies. A reduction in hospital stay was found in five of these studies.[33,34,41,45,48] Results such as these would imply a substantial saving in the cost of treatment as inpatient time is reduced. Indeed, of the three studies that looked at cost of treatment, all three found that savings were made when MDT care was introduced.[41,43,45] However, the evidence was limited as it was only presented in three studies. More studies that explore resource use to weigh up the impact of it as a collective would help to interpret the significance of resource use in this area better.

MDT and quality of life

Diabetic foot complications affect people both physically and mentally and therefore have a negative impact on a person's quality of life.[15] Quality of life was researched in just one of the included studies.[43] Although this study found the MDT had a positive impact on participants' quality of life the authors themselves reported that the questionnaire used was not without its flaws as it asked participants to draw on past events, which may have resulted in some recall bias.

With only one study reporting on quality of life it is impossible to draw firm conclusions from these results. With best practice guidelines advocating that the patient should be at the centre of their care and involved throughout their wound management process this is surprising.[5,54] Indeed, if the patient should be placed at the centre of the care plan, then it is reasonable to suggest that they should be included as a member of the team. However, none of the studies included in this review echoed this sentiment. Furthermore, a literature review that sought to explore the concept of MDTs found that the client was included as a member of the team in only 1% of the literature reviewed.[55] It would, in turn, be beneficial to see whether quality of life improves when patients are more involved in their care. It is evident, however, that further research is needed into the outcomes of diabetic foot ulceration when the service user is considered as an equal member of the team.

MDT and mortality

Mortality is a serious complication of diabetic foot ulceration and is a risk to all clients with diabetic foot disease regardless of its severity.[56] A reduction in mortality with those who were involved in MDT care was found in the two studies that addressed this outcome.[37,46] However, a breakdown of cause of death was not given, which would have been beneficial to understand the significance of this mortality rate better. Also, as only two studies reported on mortality the data are limited. Future studies in this area should strive to produce data in relation to mortality and MDT care.

Reflective Questions

- What is the significance of diabetic foot disease for the individual, the health service and society as a whole?
- Why is a multidisciplinary approach to management of the diabetic foot ulcers important?
- Why is the assessment of study quality important in determining the significance of the study results?

Quality of the evidence

Although improvements were seen in many of the outcomes post MDT care these results should be interpreted with caution. Following quality appraisal, 61% of studies included were found not to reach the required validity scores. This arose mainly due to insufficient reporting within the studies, making it many aspects of the published studies unclear. For example, many of the studies failed to give clear sample sizes. This left it impossible to interpret the results properly or to perform further statistical analysis. A further concerning point is that only one study declared that the researchers played no part in care delivery. If the researchers in any of the studies were also members of the MDT this would introduce a strong element of bias to the studies.[18]

Within this systematic review, many of the study's results were reviewed individually rather than through a meta-analysis. This was due to a substantial heterogeneity in many cases. Meta-analysis can only be performed when studies are similar to each other.[57] Furthermore, as the majority of studies were based in a hospital setting and no one study had the exact same team makeup it is difficult to generalise the results into the wider clinical setting. The evidence presented cannot be used to firmly guide future practice. From the currently available evidence, an impact of the MDT on diabetic foot outcomes can be seen but due to the lack of high-quality evidence and substantial heterogeneity these results should be interpreted with caution.

Conclusion

This systematic review aimed to explore the impact of the MDT on the management of individuals with DFUs. A comprehensive review of current evidence did find some positive effects of the MDT on DFUs, namely; amputation rate, severity of amputation and resource use. MDT care also appears to improve mortality and quality of life of people with this condition. However, due to methodological heterogeneity and some issues around methodological

© 2017 MA Healthcare ltd

Request Permissions

© MA Healthcare Ltd. Downloaded from magonlinelibrary.com by 155.198.030.043 on June 14, 2017.
Use for licensed purposes only. No other uses without permission. All rights reserved.

quality the outcomes of this review should be interpreted with caution. MDTs have been widely cited as an important strategy for managing the diabetic foot.[11,12,14] Therefore, it is important that this conclusion is interpreted as a lack of high-quality evidence rather than evidence of no effect.

The promising results found in many of the studies warrant further research. The main shortcomings of the studies included in this review are unclear reporting of sample sizes, risk of bias if the researchers were also the health professionals who were involved with the MDT, and the retrospective nature of many of the studies that may have resulted in inaccurate reporting of results. To facilitate further robust meta-analysis of the impact of diabetic foot teams, homogeneity of study design, characteristics and reporting is desirable. Addressing these issues should aim to enhance credibility of future results on this topic. **JWC**

References

1 Benbow M. Diabetic foot ulcers. Journal of Community Nursing 2012; 26(5):16–19.
2 Bakker K, Apelqvist J, Schaper NC for the International Working Group on Diabetic Foot Editorial Board. Practical guidelines on the management and prevention of the diabetic foot 2011. Diabetes Metab Res Rev 2012; 28 (Suppl 1):225–231. https://doi.org/10.1002/dmrr.2253
3 Huang D, Wilkins C, Evans D et al. The diabetic foot: the importance of coordinated care. Semin Intervent Radiol 2014; 31(04):307–312. https://doi.org/10.1055/s-0034-1393966
4 Jordan A, Graham A, Balanda K, Fahy L. Making Diabetes Count —What Does the Future Hold? A Systematic Approach to Forecasting Population Prevalence on the Island of Ireland in 2010 and 2015. The Institute of Public Health in Ireland 2007. http://bit.ly/2q8JxLG (accessed 22 May 2017).
5 Health Service Executive. National Best Practice and Evidence Based Guidelines for Wound Management. Health Service Executive, 2009 http://bit.ly/2rsFloL (accessed 22 May 2017).
6 Boulton AJ. The diabetic foot: grand overview, epidemiology and pathogenesis. Diabetes Metab Res Rev 2008; 24(S1 Suppl 1):S3–S6. Medline https://doi.org/10.1002/dmrr.833
7 Diabetes UK. Diabetes in the UK 2012: Key Statistics on Diabetes. Diabetes UK 2012. http://bit.ly/1kdggGV (accessed 22 May 2017)
8 Diabetes Expert Advisory Group, Health Service Executive. First Report: April 2008. Health Service Executive 2008. http://bit.ly/2qLKMDV (accessed 22 May 2017)
9 Dorresteijn JAN, Kriegsman DMW, Assendelft WJJ, Valk GD. Patient education for preventing diabetic foot ulceration. Cochrane Database Syst Rev 2012; Issue 10.https://doi.org/10.1002/14651858.CD001488.pub5 (accessed 25 May 2017).
10 Riley P, Rivas M, Bakker K, et al. eds. Diabetes and Foot Care: Put Feet First Prevent Amputations—Time to Act. International Diabetes Federation, International Working Group on the Diabetic Foot, 2005.
11 Turns M. The diabetic foot: an overview for community nurses. Br J Community Nurs 2012; 17(9):422–433. https://doi.org/10.12968/bjcn.2012.17.9.422
12 Hayes C. Interprofessional capacity building in diabetic foot management. Br J Nurs 2009;18(13):804–810. https://doi.org/10.12968/bjon.2009.18.13.43209
13 Korzon-Burakowska A, Dziemidok P. Diabetic foot - the need for comprehensive multidisciplinary approach. Ann Agric Environ Med 2011;18(2):314–317.
14 National Institute for Health and Care Excellence. Diabetic Foot Problems: Prevention and Management. NG 19. NICE, 2015. http://www.nice.org.uk/guidance/ng19 (accessed 22 May 2017).
15 van Acker K, Léger P, Hartemann A, Chawla A, Siddiqui MK. Burden of diabetic foot disorders, guidelines for management and disparities in implementation in Europe: a systematic literature review. Diabetes Metab Res Rev 2014;30(8):635–645. https://doi.org/10.1002/dmrr.2523
16 Kerr M. Foot Care For People With Diabetes: The Economic Case for Change. NHS Diabetes, 2012. http://bit.ly/1jCR9ho (accessed 22 May 2017)
17 The Cochrane Collaboration. Review Manager 5 (RevMan 5.3) 2014 http://tech.cochrane.org/revman/ (accessed 22 May 2017)
18 Glynn L. A critical appraisal tool for library and information research. Libr Hi Tech 2006; 24(3):387–399. https://doi.org/10.1108/07378830610692154
19 PRISMA. PRISMA 2009 Checklist. http://bit.ly/2r8EuPB (accessed 22 May 2017)
20 Thewjitcharoen Y, Krittiyawong S, Porramatikul S et al. Outcomes of hospitalized diabetic foot patients in a multi-disciplinary team—Thailand's experience. Journal of Clinical and Translational Endocrinology 2014: 1(4): 187-191. https://doi.org/10.1016/j.jcte.2014.10.002
21 Van Gils CC, Wheeler LA, Mellstrom M et al. Amputation prevention by vascular surgery and podiatry collaboration in high-risk diabetic and nondiabetic patients. The Operation Desert Foot experience. Diabetes Care 1999;22(5):678–683. https://doi.org/10.2337/diacare.22.5.678
22 Sinkin JC, Reilly M, Cralley A et al. Multidisciplinary approach to soft-tissue reconstruction of the diabetic Charcot foot. Plast Reconstr Surg 2015;135(2):611–616. https://doi.org/10.1097/PRS.0000000000000861
23 Ignatyeva VI, Severens JL, Ramos IC et al. Costs of Hospital Stay in Specialized Diabetic Foot Department in Russia. Value in Health Regional Issues 2015;7:80–86. https://doi.org/10.1016/j.vhri.2015.09.003
24 Chen MY, Huang WC, Peng YS et al. Effectiveness of a health promotion programme for farmers and fishermen with type-2 diabetes in Taiwan. J Adv Nurs 2011;67(9):2060–2067. https://doi.org/10.1111/j.1365-2648.2011.05678.x
25 Valdés AM, Angderson C, Giner JJ. A multidisciplinary, therapy-based, team approach for efficient and effective wound healing: a retrospective study. Ostomy Wound Manage 1999;45(6):30–36.
26 El Sakka K, Fassiadis N, Gambhir RP et al. An integrated care pathway to save the critically ischaemic diabetic foot. Int J Clin Pract 2006;60(6):667–669. https://doi.org/10.1111/j.1368-5031.2006.00953.x
27 Krishnan S, Nash F, Baker N, Fowler D, Rayman G. Reduction in diabetic amputations over 11 years in a defined U.K. population: benefits of multidisciplinary team work and continuous prospective audit. Diabetes Care 2008;31(1):99–101. https://doi.org/10.2337/dc07-1178
28 Manu CA, Mustafa OG, Bates M et al. Transformation of the multidisciplinary diabetic foot clinic into a multidisciplinary diabetic foot day unit: results from a service evaluation. Int J Low Extrem Wounds 2014;13(3):173–179. https://doi.org/10.1177/1534734614545877
29 Lazzarini PA, ORourke SR, Russell AW et al. Standardising practices improves clinical diabetic foot management: the Queensland Diabetic Foot Innovation Project, 2006–09. Aust Health Rev 2012;36(1):8–15. https://doi.org/10.1071/AH10978
30 Crane M, Werber B. Critical pathway approach to diabetic pedal infections in a multidisciplinary setting. J Foot Ankle Surg 1999; 38(1): 30-33.
31 Aydin K, Isildak M, Karakaya J, Gürlek A. Change in amputation predictors in diabetic foot disease: effect of multidisciplinary approach. Endocrine 2010;38(1):87–92. https://doi.org/10.1007/s12020-010-9355-z
32 Aksoy DY, Gürlek A, Çetinkaya Y et al. Change in the amputation profile in diabetic foot in a tertiary reference center: efficacy of team working. Exp Clin Endocrinol Diabetes 2004;112(9):526–530. https://doi.org/10.1055/s-2004-821310
33 Yesil S, Akinci B, Bayraktar F et al. Reduction of major amputations after starting a multidisciplinary diabetic foot care team: single centre experience from Turkey. Exp Clin Endocrinol Diabetes 2009;117(07):345–349. https://doi.org/10.1055/s-0028-1112149
34 Chiu CC, Huang CL, Weng SF et al. A multidisciplinary diabetic foot ulcer treatment programme significantly improved the outcome in patients with infected diabetic foot ulcers. J Plast Reconstr Aesthet Surg 2011;64(7):867–872. https://doi.org/10.1016/j.bjps.2010.11.024
35 Hsu CR, Chang CC, Chen YT et al. Organization of wound healing services: The impact on lowering the diabetes foot amputation rate in a ten-year review and the importance of early debridement. Diabetes Res Clin Pract 2015;109(1):77–84. https://doi.org/10.1016/j.diabres.2015.04.026
36 Rubio JA, Aragón-Sánchez J, Jiménez S et al. Reducing major lower extremity amputations after the introduction of a multidisciplinary team for the diabetic foot. Int J Low Extrem Wounds 2014;13(1):22–26. https://doi.org/10.1177/1534734614521234
37 Martínez-Gómez DA, Moreno-Carrillo MA, Campillo-Soto A et al. Reduction in diabetic amputations over 15 years in a defined Spain population. Benefits of a critical pathway approach and multidisciplinary team work. Rev Esp Quimioter 2014;27(3):170–179.
38 Meltzer DD, Pels S, Payne WG et al. Decreasing amputation rates in patients with diabetes mellitus. An outcome study. J Am Podiatr Med Assoc 2002;92(8):425–428. https://doi.org/10.7547/87507315-92-8-425
39 Armstrong DG, Bharara M, White M et al. The impact and outcomes of establishing an integrated interdisciplinary surgical team to care for

© 2017 MA Healthcare ltd

Request Permissions

© MA Healthcare Ltd. Downloaded from magonlinelibrary.com by 155.198.030.043 on June 14, 2017.
Use for licensed purposes only. No other uses without permission. All rights reserved.

the diabetic foot. Diabetes Metab Res Rev 2012;28(6):514–518. https://doi.org/10.1002/dmrr.2299
40 Williams DT, Majeed MU, Shingler G et al. A diabetic foot service established by a department of vascular surgery: an observational study. Ann Vasc Surg 2012; 26(5): 700-706. https://doi.org/10.1016/j.avsg.2011.10.020
41 Nather A, Chionh SB, Wong KL et al. Value of team approach combined with clinical pathway for diabetic foot problems: a clinical evaluation. Diabet Foot Ankle 2010;1(5731):1–5 https://doi.org/10.3402/dfa.v1i0.5731
42 Alexandrescu V, Hubermont G, Coessens V et al. Why a multidisciplinary team may represent a key factor for lowering the inferior limb loss rate in diabetic neuro-ischaemic wounds: application in a departmental institution. Acta Chir Belg 2009;109(6):694–700. https://doi.org/10.1080/00015458.2009.11680519
43 Rerkasem K, Kosachunhanun N, Tongprasert S, Guntawongwan K. A multidisciplinary diabetic foot protocol at Chiang Mai University Hospital: cost and quality of life. Int J Low Extrem Wounds 2009;8(3):153–156. https://doi.org/10.1177/1534734609344143
44 Dargis V, Pantelejeva O, Jonushaite A et al. Benefits of a multidisciplinary approach in the management of recurrent diabetic foot ulceration in Lithuania: a prospective study. Diabetes Care 1999;22(9):1428–1431. https://doi.org/10.2337/diacare.22.9.1428
45 Nason GJ, Strapp H, Kiernan C et al. The cost utility of a multi-disciplinary foot protection clinic (MDFPC) in an Irish hospital setting. Ir J Med Sci 2013;182(1):41–45. https://doi.org/10.1007/s11845-012-0823-8
46 Cahn A, Elishuv O, Olshtain-Pops K. Establishing a multidisciplinary diabetic foot team in a large tertiary hospital: a workshop. Diabetes Metab Res Rev 2014;30(5):350–353. https://doi.org/10.1002/dmrr.2527
47 Hedetoft C, Rasmussen A, Fabrin J, Kølendorf K. Four-fold increase in foot ulcers in type 2 diabetic subjects without an increase in major amputations by a multidisciplinary setting. Diabetes Res Clin Pract 2009;83(3):353–357. https://doi.org/10.1016/j.diabres.2008.11.025
48 Larsson J, Stenström A, Apelqvist J, Agardh CD. Decreasing incidence of major amputation in diabetic patients: a consequence of a multidisciplinary foot care team approach? Diabet Med 1995;12(9):770–776. https://doi.org/10.1111/j.1464-5491.1995.tb02078.x
49 Quinlivan E, Jones S, Causby R, Brown D. Reduction of amputation rates in multidisciplinary foot clinics—a systematic review. Wound Practice and Research. 2014;22(3):155–162. http://bit.ly/2rt2fBA
50 Higgins JP, Green S, eds. Cochrane Handbook for Systematic Reviews of Interventions Version 5.1.0 [updated March 2011]. The Cochrane Collaboration, 2011. http://handbook.cochrane.org/ (accessed 25 May 2017).
51 Liberati A, Altman DG, Tetzlaff J et al. The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate healthcare interventions: explanation and elaboration. BMJ 2009;339:b2700. https://doi.org/10.1136/bmj.b2700
52 National Diabetes Programme Clinical Strategy and Programmes Directorate, Health Service Executive. Model of Care for the Diabetic Foot. Health Service Executive, 2011; http://bit.ly/1mpv1qc. (accessed 22 May 2017).
53 Yazdanpanah L, Nasiri M, Adarvishi S. Literature review on the management of diabetic foot ulcer. World Journal of Diabetes 2015;6(1):37–53. https://doi.org/10.4239/wjd.v6.i1.37
54 Quality and Patient Safety Directorate, Health Service Executive. Quality and patient safety: clinical governance development ... an assurance check for health service providers. Health Service Executive 2012. http://bit.ly/2qbMlac (accessed 22 May 2017).
55 Moore Z, Butcher G, Corbett L et al. Exploring the concept of a team approach to wound care: managing wounds as a team. AAWC, AWMA, EWMA Position Paper. http://bit.ly/2qGTC5l (accessed 22 May 2017)
56 Aveyard H. Doing A Literature Review in Health and Social Care: A Practical Guide. Open University Press, 2007.
57 Moulik PK, Mtonga R, Gill GV. Amputation and mortality in new-onset diabetic foot ulcers stratified by etiology. Diabetes Care 2003;26(2):491–494. https://doi.org/10.2337/diacare.26.2.491

JWC available at www.magonlinelibrary.com

Register for free and take advantage of these great benefits and features:

- Save JWC articles for quick access
- Save your searches: great if you frequently search for the same criteria
- Get citation alerts to track citations to specific articles

Once you've registered, you can sign up for table of contents alerts for JWC, delivered to you by email as soon as new content becomes available. Simply visit the journal and click 'TOC Alerts'.

www.magonlinelibrary.com/r/jowc

© 2017 MA Healthcare ltd

Request Permissions

© MA Healthcare Ltd. Downloaded from magonlinelibrary.com by 155.198.030.043 on June 14, 2017.
Use for licensed purposes only. No other uses without permission. All rights reserved.